



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 • STANDARD 6.AT.8

# Equations and Inequalities Problem Solving

Lesson 7-7 • Teacher Copy

**Name:** \_\_\_\_\_ **Date:** \_\_\_\_\_ **Class:** \_\_\_\_\_



## TODAY'S OBJECTIVES

**Content Objective:** I can model and solve real-world problems using equations and inequalities.

**Language Objective:** I can explain my reasoning using the words model, equation, inequality, and reasonableness.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Model

Equation

Inequality

Reasonableness

Variable

Constraint



Tap any word to see what it means and a picture.

1 A drawing, equation, or diagram that represents a situation is a

2 A math sentence with an equal sign is an .

3 A math sentence that compares values with  $<$ ,  $>$ ,  $\leq$ , or  $\geq$  is an

4 Checking whether an answer makes sense for the problem is checking its

5 A letter that stands for the unknown amount in a problem is a

6

A limit that tells which values are allowed in a problem is a

## Reflect — Exit Ticket

A box of donuts has some donuts. After giving away 7, there are fewer than 5 left. Which inequality represents the starting number of donuts  $d$ ?

A.  $d - 7 < 5$

B.  $d + 7 < 5$

C.  $d - 7 > 5$

D.  $d - 7 = 5$

Your answer:

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## Answer Key & Teacher Guide

1. **Notes 1:** Model
2. **Notes 2:** Equation
3. **Notes 3:** Inequality
4. **Notes 4:** Reasonableness
5. **Notes 5:** Variable
6. **Notes 6:** Constraint
7. **Watch for this mistake:** *A common mistake in Equations and Inequalities Problem Solving is skipping the key idea: "'Equals' or 'exactly' points to an equation ( $=$ ); 'at least, at most, more than, fewer than' points to an inequality." — always check your work against this rule before you submit.*
8. **Try It 1:** A.  $x \leq 20$  — 'At most 20' means 20 or fewer, so  $x \leq 20$ .
9. **Try It 2:** A.  $w = 12$  — Divide both sides by 5:  $w = 60 \div 5 = 12$  pounds. Check:  $5 \times 12 = 60$  ✓
10. **Exit Ticket:** A.  $d - 7 < 5$  — Starting with  $d$  donuts and giving away 7 leaves fewer than 5:  $d - 7 < 5$ .  
Solve:  $d < 12$ . ✓